

Relationship between the prenatal exposure to low-level of mercury and the size of a newborn's cerebellum.



Exposure to methylmercury at any stage of central nervous system development could induce alterations and result in severe congenital abnormalities. Total mercury level in maternal hair during pregnancy correlates well with blood levels of methylmercury and with total mercury levels in fetal brain. A prospective study has been conducted and a total of 137 childbearing women living at the coastal region with term, normal pregnancies were included and their newborns evaluated by ultrasonography. Mothers and their newborns are divided in two groups according to their hair mercury levels; examined group with high body levels of mercury ($\geq 1 \mu\text{g/g}$) and control group with low body levels of mercury ($<1 \mu\text{g/g}$). Neurosonographic examination was conducted to all newborns. Two dimensions of cerebellum in the sagittal-medial plane have been measured: maximum height and width starting from the roof of the fourth chamber. Majority of mothers had hair mercury levels lower than $1 \mu\text{g/g}$ ($N = 107$). Mean value was $0.88 \mu\text{g/g}$ (SD 1.24), ranging from 0.02 to $8.71 \mu\text{g/g}$. There was no significant difference between the two groups when it comes to the width of cerebellum (Mann-Whitney test: $Z = 1471$; $p = 0.141$). However, comparison related to the length of cerebellum shows statistically significant smaller cerebellum in newborns whose mother had hair mercury levels higher than $1 \mu\text{g/g}$ (Mann-Whitney test: $Z = 2329$; $p = 0.019$). Our results lead to a conclusion that prenatal exposure to, what we consider to be, low-levels of methylmercury does influence fetal brain development detected as decreased size of newborn's cerebellum. From a clinical point of view, a question related to the influence of prenatal low-level methylmercury exposure on fetal neurodevelopment remains open. Our further objectives are to direct the research towards performing detailed neuropsychological tests on children at the age of 18 months. Such tests could indicate the presence of subtle neurological or neuropsychological deficits.

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